

Trainings ▾

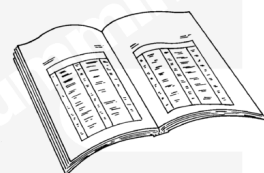
Preparatory Steps

QSK45 Engines

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

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ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift this component.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>)
- Remove the coolant filter head. Refer to Procedure 008-007 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-007-tr.html>)
- Remove the thermostat housing. Refer to Procedure 008-015 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-015-tr.html>)

with Mechanically Actuated Injector

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- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>)
- Remove the front intercooler. Refer to Procedure 010-083 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-010-083-tr.html>)
- Remove the coolant filter head. Refer to Procedure 008-007 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-007-tr.html>)
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- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>)

QSK60 Marine Applications

** WARNING **

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

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- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>)
- Remove the water outlet manifold. Refer to Procedure 008-061 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-061-tr.html>)
- Remove the front intercooler. Refer to Procedure 010-083 in Section 10. (</qs3/pubsys2/xml/en/procedures/56/56->

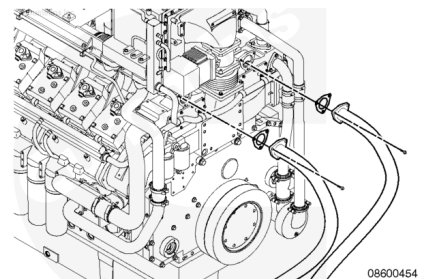
010-083-tr.html)

- Remove the sea water expansion tank. Refer to Procedure 008-052 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-052-tr.html)
- Remove the sea water heat exchanger. Refer to Procedure 008-053 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-053-tr.html)
- Remove the coolant filter head. Refer to Procedure 008-007 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-007-tr.html)
- Remove the thermostat housing. Refer to Procedure 008-015 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-015-tr.html)

Remove

with Mechanically Actuated Injector

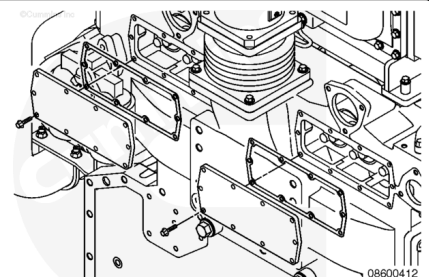
Remove the upper radiator and aftercooler hoses on the thermostat housing.



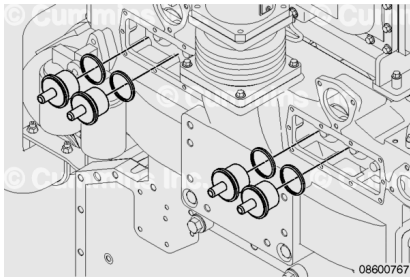
Remove the twenty capscrews holding the thermostat housing covers on the right and left side of the thermostat housing.

Remove both thermostat housing covers and discard both thermostat housing cover gaskets.

Note : With the thermostat housing covers removed, it exposes three thermostats on the left and three thermostats on the right. One thermostat on the left side is mounted upright for the LTA cooling system.



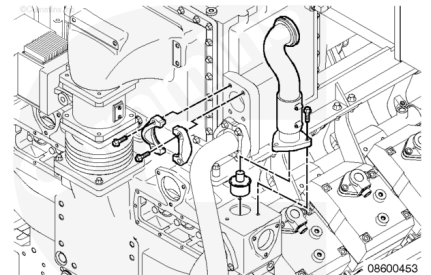
Remove the thermostats.



LTA

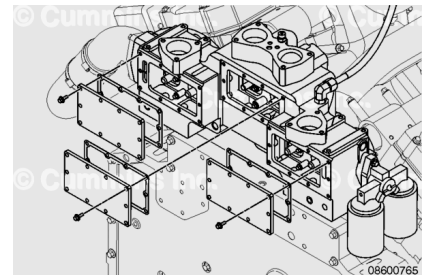
The low temperature aftercooler (LTA) thermostat is accessed by removing the hose and outlet connection from the left thermostat housing.

Remove the low temperature aftercooler (LTA) thermostat.



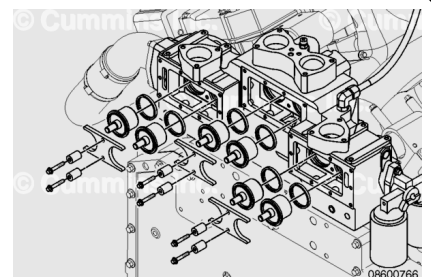
with Electronically Actuated Injector

Remove the thermostat cover plates. Each cover plate has ten capscrews.



Remove the two capscrews in each clamping plate and remove the spacers and clamping plates.

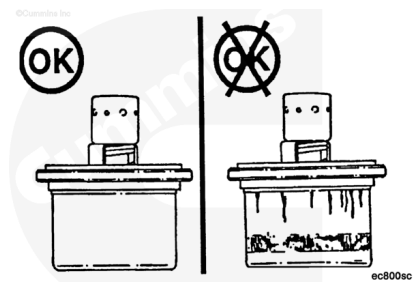
Remove the thermostats and seals from the housing.



Inspect for Reuse

The two center thermostats are low temperature aftercooling thermostats and have a different temperature range than the engine thermostats. They are knurled on the upper diameter.

Check the thermostats for wear or damage. If the barrel of a thermostat is worn or fretted, it **must** be replaced.

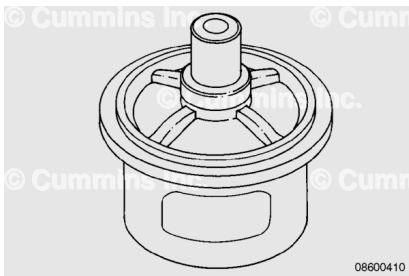


Test

Check the thermostat marking to see when the thermostat is designed to open.

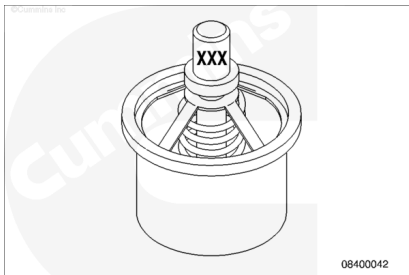
The rated nominal temperature is marked on the side of the thermostat.

The design temperature of the thermostat is 82°C [180°F].



Suspend the thermostat and thermometer in a container of water. Do **not** allow the thermostat or thermometer to touch the side of the container.

Heat the water to check the temperature when the thermostat begins to open.



Thermostat Initial Opening Temperature

celsius		fahrenheit
74.4	MIN	166
78.3	MAX	173

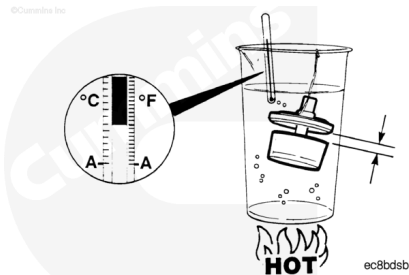
Continue to apply heat to check the temperature when the thermostat is completely open.

Minimum Open Distance at Full Open Temperature

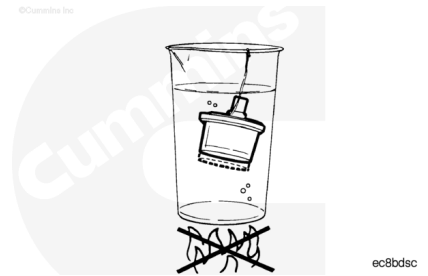
mm		in
10.0	MIN	0.40

Full-Open Temperature

celsius		fahrenheit
90	MIN	194



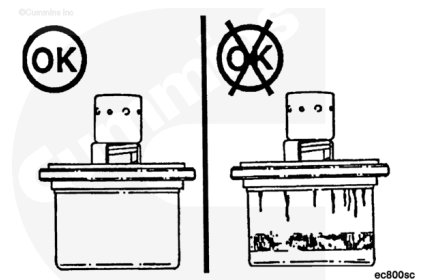
Remove the container from the heat.



With the thermostat open, check for wear or damage.

If the seat of the thermostat is worn or fretted, it **must** be replaced.

Make sure the thermostat closes when the temperature is below 74.4°C [166°F].

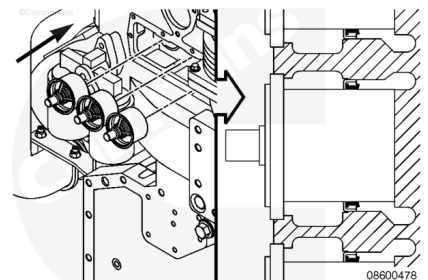


Install

with Mechanically Actuated Injector

Install the thermostat seals. Refer to Procedure 008-016 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-016.html>)

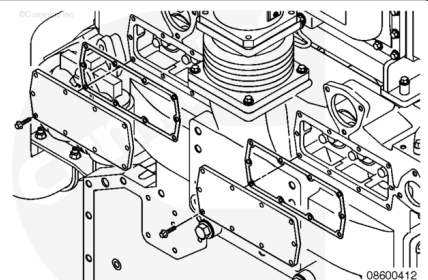
Install the thermostat by pushing on the outer rim.



Install the thermostat housing cover gasket, right and left covers, and capscrews.

Tighten the capscrews.

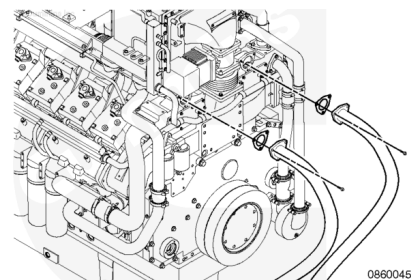
Torque Value: 45 n•m [33 ft-lb]



Install the upper radiator and aftercooler hoses on the thermostat housing.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



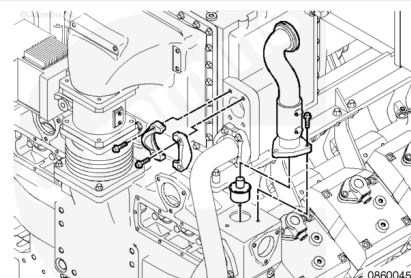
LTA

Install the LTA thermostat.

Install the outlet hose and connection.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



with Electronically Actuated Injector

Note : The two center thermostats are the low temperature aftercooler thermostats. The outermost thermostats are for the engine jacket water cooling.

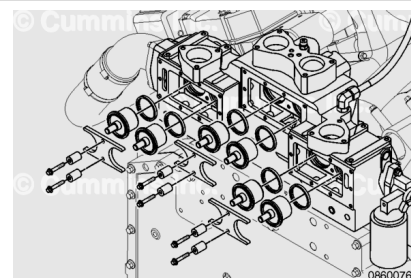
Install the thermostat seals. Refer to Procedure 008-016 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-016.html)

Install the thermostat by pushing on the outer rim.

Install the clamping plates, spacers and capscrews.

Tighten the capscrews.

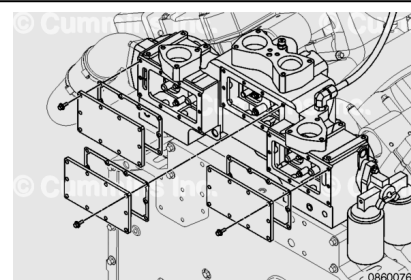
Torque Value: 25 n•m [221 in-lb]



Install the thermostat housing cover gaskets, covers, and capscrews.

Tighten the capscrews.

Torque Value: 25 n•m [221 in-lb]

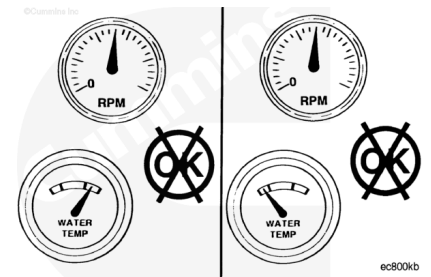


Leak Test

Note : This test procedure does **not** apply to thermostat housings with a vent hole from the inlet side to the full flow passage chamber.

The engine thermostat and thermostat seal **must** operate properly in order for the engine to operate in the most efficient heat range. Overheating or overcooling will shorten engine life.

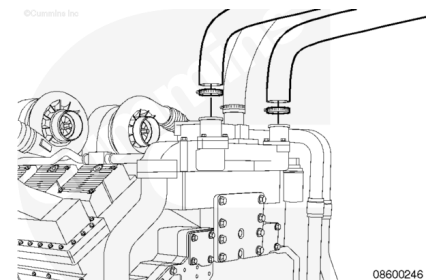
With a low temperature aftercooling thermostat, the intake air temperature is monitored by the electronic system and will generate a fault code when the intake air temperature exceeds the maximum temperature limits.



Engine Coolant Temperature

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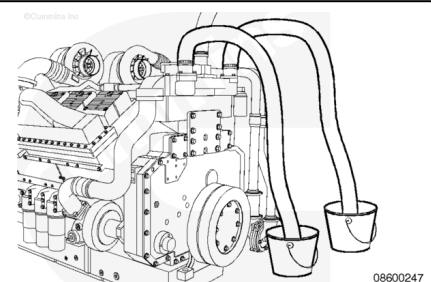
Note : If the engine being tested has low temperature aftercooling, leave the low-temperature aftercooling radiator hoses connected to the thermostat housing.

Remove the upper engine radiator hoses from the thermostat housing.

Install two hoses of the same size on the thermostat housing outlets. The hoses **must** be long enough to reach a remote, dry container.

Install and tighten hose clamps on the housing outlets.

Put the end of the hoses in the two dry containers.

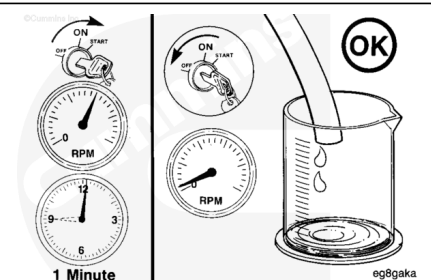


Operate the engine at rated rpm for 1 minute.

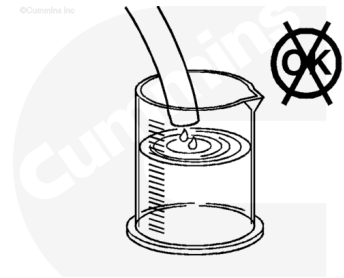
Shut the engine off.

Measure the amount of coolant collected in each container.

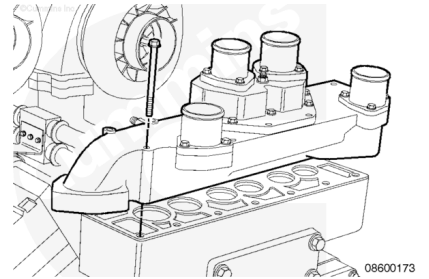
The amount of coolant collected **must not** be more than 100 cc [3.3 fl oz].



If more than 100 cc [3.3 fl oz] of coolant is collected, the thermostat or the thermostat seal is leaking.

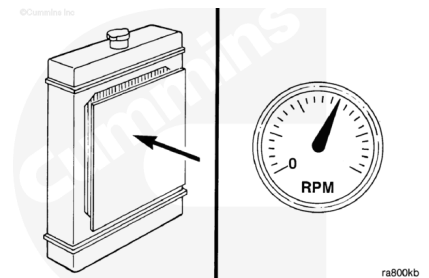


Remove the thermostats and replace the seals. Refer to Procedure 008-015 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-015-tr.html)
Test the thermostats for proper operation.



High Coolant Temperature

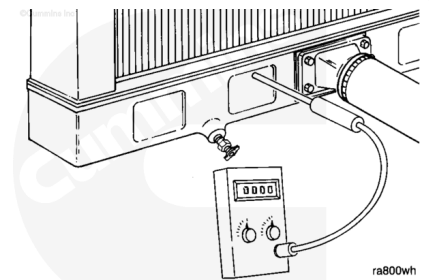
Restrict the radiator airflow. Operate the engine until the coolant temperature reaches 90 to 93°C [194 to 200°F].



Use a contact pyrometer (shown), or install a temperature gauge in the desired locations.

Record the temperature of the coolant outlet hoses. An increase in temperature indicates the thermostats have started to open.

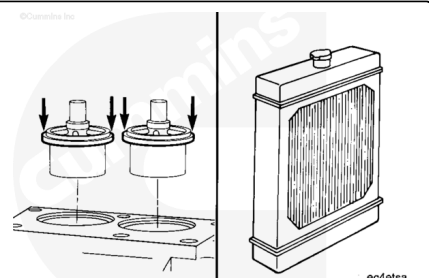
Record the temperature of the radiator bottom tank or coolant out tube.



If the difference in temperature is more than 8°C or 15°F, either the thermostats are **not** fully open or the radiator core is plugged.

Check the radiator.

Refer to Procedure 008-042 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-042.html)



If the radiator is in good working order, replace the thermostats.

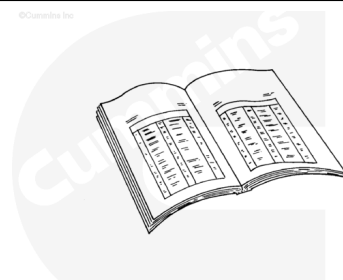
Finishing Steps

QSK45 Engines

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ck800wa

- Install the thermostat housing. Refer to Procedure 008-015 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-015-tr.html>)
- Install the coolant filter head. Refer to Procedure 008-007 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-007-tr.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>)
- Operate the engine until the coolant temperature reaches 70°C [158°F]. Check for leaks.

with Mechanically Actuated Injector

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- Install the front intercooler. Refer to Procedure 010-083 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-010-083-tr.html>)
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- Operate the engine until the coolant temperature reaches 70°C [158°F]. Check for leaks.

QSK60 Marine Applications

WARNING

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- Install the coolant filter head. Refer to Procedure 008-007 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-007-tr.html>)
- Install the sea water heat exchanger. Refer to Procedure 008-053 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-053-tr.html>)
- Install the sea water expansion tank. Refer to Procedure 008-052 in Section 8.

(/qs3/pubsys2/xml/en/procedures/56/56-008-052-tr.html)

- Install the front intercooler. Refer to Procedure 010-083 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-083-tr.html)
- Install the water outlet manifold. Refer to Procedure 008-061 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-061-tr.html)
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